



Aida Farahani, PhD

AI Solutions Engineer · Industrial AI · Computer Vision

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ABOUT

I am an AI Solutions Engineer with a PhD in Artificial Intelligence, focused on building practical, reliable AI systems for real-world applications.

My work centers on industrial computer vision, data-centric AI, and end-to-end system design: from problem definition and dataset construction to model development, evaluation, integration, and deployment.

I specialize in improving model performance through data refinement and in designing hybrid systems that combine vision, geometry, and language models.

PhD in Artificial Intelligence — Technical University of Chemnitz

Applied AI Focus — Industrial computer vision, data-centric AI, and production-oriented workflows

Applied & Scientific Foundation — Practical AI for real-world systems, grounded in scientific machine learning and research methods

CORE EXPERTISE

Computer Vision — Object detection, instance segmentation, industrial image understanding, field-condition evaluation, and visual inspection workflows.

Data-Centric AI — Dataset design, annotation strategy, class balancing, failure-case analysis, and iterative model improvement.

AI Systems — End-to-end pipelines, workflow integration, production-oriented design, validation logic, and deployment planning.

Multimodal AI — Vision-language workflows: semantic retrieval, document/PDF understanding, and structured extraction from mixed media.

Deployment — CoreML, Swift/iOS integration, on-device inference, model export, and mobile visualization prototypes.

Scientific ML — FEM surrogate modeling, 3D deformation learning, mesh autoencoders, implicit representations, and RL for process modeling.

SKILLS

Python libraries — pandas, trimesh, plotly, meshlab, PyTorch Geometric; mesh autoencoders, SDF, and FEM data workflows.

Frameworks — PyTorch, TensorFlow, scikit-learn; computer vision (detection, segmentation, industrial imagery, preprocessing, evaluation).

ML Tools — Docker, conda, Linux, Git, Jenkins, Jupyter; RAG, CLIP, document/PDF understanding, Ollama, Unsloth.

Programming & deployment — C#.NET, C++, MATLAB, Python; CoreML, Swift/iOS, on-device inference, model export.

SELECTED PROJECTS

Multimodal Image Search

Retrieval · 2025

Built semantic image retrieval tools using language-driven search and similarity-based clustering.

Focus: CLIP-based embeddings, multimodal retrieval, image search, and clustering workflows.

Document & PDF Understanding

Vision-Language · 2025

Developed pipelines for extracting structured data from documents using vision-language methods.

Focus: document understanding, structured extraction, multimodal parsing, and workflow automation.

Data-Centric Model Improvement

Model Reliability · 2024 – present

Designed workflows to improve model performance through targeted data quality improvements.

Focus: annotation strategy, failure-case driven dataset updates, and iterative performance optimization.

Industrial Vision: Trench & Fiber Detection

Industrial AI · 2024

Designed segmentation pipelines for detecting trenches, ducts, and fittings in noisy field conditions.

Focus: instance segmentation for complex scenes, dataset refinement, class balancing, and field-condition evaluation.

On-Device Vision Inference

Deployment · 2024

Implemented real-time vision models on mobile devices with model optimization and iOS integration.

Focus: CoreML export, Swift/iOS integration, visualization, and field-ready inference prototypes.

Smart Airsense (BMBF)

Jan 2022 – 2024

Researcher on the BMBF project investigating AI methods for an interactive health assistant based on human-in-the-loop machine learning.

Focus: time series classification using air particle sensor measurements to predict environmental events.

Partners: Corant GmbH, TU Chemnitz.

3D Deformation Modeling (FEM Surrogate)

PhD Research · 2021 – 2023

Used reinforcement learning approaches for reverse engineering of metal forming process chains (prediction of actions for a desired deformed metal part).

Focus: sequential deformation modeling and reinforcement learning for process optimization.

ML@Karoprod (BMBF)

Feb 2018 – Jun 2021

Researcher on the BMBF project applying machine learning for process and quality prediction in automotive body production.

Focus: FEM-based dataset generation, deformation-property prediction, reinforcement learning for process modeling, and simulation-to-ML workflows for forming processes.

Partners: Fraunhofer IWU Dresden, Scale GmbH, TU Chemnitz.

Project code: hamkerlab.github.io/ML-Karoprod-MeshPredictor

RESEARCH & EDUCATION

PhD in Artificial Intelligence

Technical University of Chemnitz, Germany · 2026

Thesis: *Exploring Deep Learning Approaches for 3D Deformation : Toward Finite Element Method Distillation*

M.Sc. Mechatronics Engineering

K.N. Toosi University of Technology, Iran · 2010

Thesis: *Visual servoing and object pose estimation*

B.Sc. Computer Software Engineering

IAU Tehran Branch, Iran · 2006

Thesis: *Persian handwriting recognition*

LANGUAGES

- ▶ **Farsi** — native
- ▶ **English** — business fluent
- ▶ **German** — B2 (upper-intermediate, professional working proficiency)
- ▶ **French** — basic